R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

First YEAR (Semester – I

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Advanced Mathematics and Numerical Analysis**	**MTSE-1001**	**3L-1T-0P**	**4**

Course Objective:

CO1:-To solve the numerical solution of partial differential equation give tool and technic to solve numerical of various problem pertaining to hyperbolic, parabolic and elliptic types by finite difference method.

CO2:- To solve the real life application of transforms to boundary value problem in engineering by integral transform

CO3:- To solving Integral equations for Conversion of Linear Differential equation (LDE) to an integral equation (IE) which applicable for solving real life problems.

CO4:- Application of calculus of variation to various engineering problems

CO5:- To solve various problems in the field of engineering by finite elements

Syllabus:

Unit 1:

Numerical solution of Partial Differential Equation (PDE): Numerical solution of PDE of hyperbolic, parabolic and elliptic types by finite difference method.

Unit 2:

Integral transforms: general definition, introduction to Mellin, Hankel and Fourier transforms and fast Fourier transforms application of transforms to boundary value problems in engineering

Unit 3:

Integral equations: Conversion of Linear Differential equation (LDE) to an integral equation (IE), conversion of boundary value problems to integral equations using Greenns function, solution of Integral equation, IE of convolution type, Abelns IE, Integral differential equations, IE with separable variable, solution of Freehold Equation with separable kernels, solution of Freehold and Volterra equations by method of successive approximations.

Unit 4:

Calculus of Variation: Functional and their Variation, Eulerns equation for function of one and two independent variables, application to engineering problems.

Unit 5:

FEM: Variation functional, Euler Lagrangens equation, Variation forms, Ritz methods, Galerkinns method, descretization, finite elements method for one dimensional problem

Course Outcomes:

- On completion of this course the students will be able to solve various problems in the field of engineering employing probability and statistical methods.

Reference Books:

Reference Books

1. B.V. Rammana, Higher Engineering Mathematics, Tata McGraw Hill Pub. Company, 2007.
2. Potter, Goldberg & Edward, Advanced Engineering Mathematics, Oxford University Press.
3. S. S. Shastry, Engineering Mathematics, PHI Learning
4. C.B. Gupta, Engineering Mathematics I & II, McGraw Hill India, 2015
5. Dean G. Duffy, Advanced Engineering Mathematics with MATLAB, CRC Press, 2013.
6. E. Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons Inc.
7. S.D. Sharma & Kedarnath, Operations Research, Ramnath & Co.
8. M. K. Jain, Iyengar, Numerical Methods for Scientific and Engineering Computation, New Age

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

First YEAR (Semester – I

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Theory of Elasticity**	**MTSE--1002**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the concept of plane stress and strain and differential equations of equilibrium in solids.

CO2: Analyze the two-dimensional problems in rectangular co-ordinates.

CO3: Analyze the two-dimensional problems in polar co-ordinates

CO4: Analyze the stress and strain in three-dimensional problems.

CO5: Analyze the various sections of prismatic bars subjected to torsion.

Syllabus:

Unit 1:

Plane Stress & Plane Strain: Plane Stress, Plane Strain, Stress and Strain at a points, Differential equations of equilibrium, constitutive relation : anisotropic materials Linear elasticity; Stress, strain, constitutive relations; Boundary conditions, Compatibility equation, stress function.

Unit 2:

Two Dimensional Problems in Rectangular Co-ordinates: Solutions by Polynomials, Saint-Venantns Principle, Determination of displacements, bending of beams, solution of two-dimensional problem in Fourier series.

Unit 3:

Two Dimensional Problems in Polar Coordinates : General equations in Polar coordinates, Pure bending of curved bars, displacements for symmetrical stress distributions, bending of curved bar, stress distribution in plates with circular holes, stresses in a circular disc general solution.

Unit 4:

Analysis of stress and strain in Three Dimensions : Principal stress and strain, shearing stress and strains, elementary equation of equilibrium , compatibility conditions, problems of elasticity involving pure bending of prismatic bars.

Unit 5:

Torsion of Prismatic Bars : Torsion of prismatic bars, membrane analogy, torsion of a bar of narrow rectangular cross section, torsion of rectangular bars, solution of torsional problem, torsion of rolled section, torsion of hollow shafts and thin tubes, torsion buckling torsional flexural buckling.

Course outcomes:

- On completion of this course the students will be familiar to the concept of elastic analysis of plane stress and plane strain problems, beams on elastic foundation and torsion on noncircular section.

References Books

1. Timoshenko, S.P. Theory of Elasticity (7th ed.). McGraw –Hill International.

2. Timoshenko, S. P. & Gere, James M. Theory of Elastic Stability (2nd. ed.). McGraw –Hill International Editions, Mechanical Engineering Series.

3. Iyenger, N.G.R. Structural Stability of Columns & Plates. Ellis Horwood Ltd, Publisher

4. Irving Shames, Advanced Solid Mechanics,

5. Popov and Balan, Mechanics of Solids

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

First YEAR (**Semester – I**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Advance Structural Analysis**	**MTSE--1003**	**3L-1T-0P**	**4**

Course Objectives:

After studying this subject, the students will be able to:

CO1: Understand the various methods methods of analysis of determinate and indeterminate structures.

CO2: Understand the matrix concept and various matrices approach for analysis of structures.

CO3: Analyze the plane truss and continuous beams using flexibility method.

CO4: Analyze the plane truss and continuous beams using stiffness method.

CO5: Analyze the plane and space frames, grids using flexibility and stiffness method.

Syllabus:

Unit 1 Review of Basic Methods of Structural Analysis

Structural elements, joints and supports, Stability, Rigidity, Static indeterminacy, Kinematic indeterminacy, Direct and indirect loading, Equilibrium, Compatibility, Force-displacement relations; Analysis of statically determinate structures (trusses, beams, frames), Principle of virtual work, Displacement-based and force-based energy principles.

Unit 2 Matrix Concepts and Matrix Analysis of Structures

Matrix, Vector, Basic matrix operations, rank, Solution of linear simultaneous equations; eigenvalues and eigenvectors.

Introduction, Coordinate systems, Local and global coordinates, displacement and force transformation matrices, Contra-gradient principle; element and structure stiffness matrices, Element and structure flexibility matrices, Equivalent joint loads, Stiffness and flexibility approaches, Evaluation of flexibility and stiffness coefficients, Relationship between flexibility and stiffness methods.

Unit 3 Matrix Method (Flexibility/ Force Method)

Analysis of plane truss and continuous beam by flexibility methods, Energy approach in flexibility method, Effect of support displacement and transformation.

Unit 4 Matrix Method (Stiffness/ Displacement Method)

Analysis of plane truss and continuous beam, by stiffness methods, Energy approach in stiffness method, Approach for global stiffness matrix, Effect of support displacement and transformation.

Unit 5 Matrix analysis of plane and space frames, grids

Symmetrical & anti-symmetrical problems, Analysis of plane and space frames with pin joints and rigid joints and grids by flexibility and stiffness methods

Course outcomes:

On completion of this course students will be able to analyse reinforced concrete beams for strength and deformation behaviour. They will be able to test dynamic testing on steel beams, static cyclic load testing of reinforced concrete frames and non-destruction testing on concrete.

Reference Books:

1. Reddy C. S., Basic Structural Analysis, Tata Mc-Graw Hill,
2. Wearer Jr. W. and Gere J. M., Matrix Analysis of Framed Structures, CBS Publications.
3. Rajsekeran, Sankarsubramanian, Computational structural Mechanics, PHI
4. Pandit G. S. and Gupta S. P., Structural Analysis: A Matrix Approach, Tata Mc-Graw Hill Pvt. Ltd., New Delhi.
5. Jain A. K., Advanced Structural Analysis, Nem Chand Brothers.
6. Kanchi M. B., Matrix Structural Analysis, John Willey Publishers
7. Meek J. L., Matrix Methods of Structural Analysis, Mc-Graw Hill
8. Menon Devdas, Advanced Structural Analysis, Narosa Publishing House, 2009.
9. Aslam Kasmali, Matrix Analysis of Structures, Brooks/Cole Publishing Co., USA, 1999.
10. Ghali A., Neville A. M. and Brown T. G., Structural Analysis: A Unified Classical and Matrix Approach, Chapman & Hall, Sixth Edition, 2007

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

First YEAR (Semester – I

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Advance Designs of Concrete Structures**	**MTSE--1004**	**3L-1T-0P**	**4**

Course Objectives

After studying this subject, the students will be able to:

CO1: Design and draft detailed drawings of the multi-story buildings and shear walls

CO2: Design and draft detailed drawings of corbels, deep beams, grid floors, column supported slabs.

CO3: Design and draft detailed drawing of combined footing, Raft/Mat foundation and Pile foundation

CO4: Design and draft detailed drawings of ground supported, over-head tanks of various shapes like rectangular, circular and Intze type.

CO5: Design and draft detailed drawings of square, rectangular and circular bunkers, silos and towers subjected to wind load.

Syllabus:

Course Contents

Unit 1: Design of multistorey buildings

Design of multistorey buildings, Ductile detailing of RCC frames for seismic forces, Design of Shear walls; Design examples.

Unit 2: Design of Special R.C. Elements

Design of Corbels, Design of Deep beams, Design of Grid floors, Design of column supported slabs (with/without beams) under gravity loading; Design examples.

Unit 3: Design of Foundations

Design of Combined footing, Raft/Mat foundation and Pile foundation; Design examples.

Unit 4: Design of Structures for storage of liquids

Design of water tanks resting on ground, Underground and Over head tanks, Design of water tanks of various shapes like rectangular, circular and Intze type tanks, Details of IS 3370 (2009), Design examples.

Unit 5: Design of Silos, Bunkers and Towers

Design of square and rectangular bunkers, Design of circular bunkers, Design of Silos;

Design examples.

Design of Towers - Computation of moments due to wind loads, Wind load analysis of a tower with circular group of columns; Design examples.

Course outcomes:

On completion of this course the students will have the confidence to design various concrete structures and structural elements by limit state design and detail the same for ductility as per code requirements.

Reference Books:

1. Varghese, P.C. (2009), Advanced RCC Designs (2nd ed.)/latest edition Prentice-Hall of India Private Limited.
2. Shah and Karve (2003), Text book of Reinforced Concrete. Structures Publications.
3. Punmia, B.C., Jain, Ashok Kumar & Jain, Arun Kumar, RCC Design, Laxmi Publications.
4. N. Krishna Raju - Advanced R.C.C. Design, 2nd / latest edition, CBS Publishers New Delhi.
5. Noel Everard (1987), Theory and Problems of RCC Design, Shaum's Outline Series (2nded.)/latest edition McGraw-hill Professional.
6. Pankaj Agrawal and Manish Shrikhande, Earthquake Resistance Design of Structures, Prentice-Hall of India Private Limited New Delhi.
7. Chandrasekaran, Jai Krishna (1994), Elements of Earthquake Engineering (2nd ed.)/latest edition South Asian Publishers.

IS codes: Latest versions

1. IS: 456 (2000) Code of practice for plain and reinforced concrete, Bureau of Indian Standards, New Delhi.
2. IS: 875 (1987) Code of practice for design loads (other than earthquake) for buildings and structures, Bureau of Indian Standards, New Delhi.
3. IS: 1893 (2016) Criteria for earthquake resistant design of structures, Bureau of Indian Standards, New Delhi.
4. IS: 3370 (2009) Concrete Structures for storage of liquids, Bureau of Indian Standards, New Delhi.
5. IS: 13920 (2016) Ductile Design and Detailing of R.C.C. structures, subjected to Seismic forces, Bureau of Indian Standards, New Delhi.

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

First YEAR (**Semester – I**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Analysis of Deep Foundation**	**MTSE—1005(A)**	**3L-1T-0P**	**4**

Course Objectives

After studying this subject, the students will be able to:

CO1: Student will have a deep understand Functions and requisites of a pile foundation

CO2: Student will have a deep understand load carrying capacity of piles by static load by IS Code methods and other behaviour

CO3: Student will have a deep understand load carrying capacity of piles by static load by Pile driving formulae and other behaviour

CO4:. Student will have a deep understand load Group action in piled foundations

CO5: Student will have a deep understand Pile subjected to lateral load and behaviour.

Unit1:-Functions and requisites of a foundation - Different types - Choice of foundation type – Types of deep foundation – Types of pile foundations- Factor governing choice of type of pile –Choice of pile materials.

Unit-2

Load carrying capacity of piles by static formulae- Introduction: IS code method- API method-Piles in cohesive and cohesion less soils – Piles in layered cohesive and cohesionless soils – Settlement of single pile – Piles bearing on rock – Piles in fill and Negative skin friction.

Unit-3

Load carrying capacity of piles by static formulae: Introduction- Pile driving formulae- selection of pile hammers- Determination of temporary elastic compression- Driving stresses in piles- Field measurement- Wave equation analysis.

Unit-4

Group action in piled foundations: Introduction- Minimum spacing of piles- group efficiency-Estimation of group bearing capacity- Effect of pile arrangement- Effect on pile groups of installation methods- precaution against heave effect in pile group-Settlement of pile group- Reduce differential settlement in pile group.

Unit-5

Pile subjected to lateral load: Introduction- Lateral resistance of single pile-IS 2911 method for lateral resistance of pile- Broms charts for lateral load analysis-Elastic analysis-p-y curves, use of p-y curves- improving lateral resistance of piles- field test on piles.

References

1. J.E. Bowles, “Foundation Analysis and Design”, McGraw Hill, 1996.
2. M.J. Tomlinson, “Foundation Design and Construction”, Addison Wesley, 2001.
3. M.J. Tomlinson, “Pile Design and Construction Practice”, E & FN Spon, 1987.
4. Braja M. Das., “Principles of Foundation Engineering”, Thomson Asia Pte , 1987,
London Ltd., Singapore, 2005, A viewpoint publication.
5. P.C. Varghese, “Foundation Engineering”, Prentice-Hall of India, New Delhi, 2005.

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

First YEAR (Semester – I

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Theory of plates and shell**	**MTSE—1005(B)**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the concept of various plate analysis theories.

CO2: Analyze rectangular plates subjected to various loadings

CO3: Analyze circular plates subjected to various loadings

CO4: Understand the concept of various shell analysis theories

CO5: Analyze shells under different loads.

Course Contents

Unit 1

Introduction to Plate Theory, Assumptions made in the Poission-Kirchoff plate theory, Plate equation and behavior of thin plates in Cartesian coordinates

Unit 2

Analysis of Rectangular Plates Subjected to various loading, Navier's method of solution for simply supported plates, Leavy's method of solution for plates under different boundary conditions.

Unit 3

Analysis of Circular Plates Circular plates, governing differential equation in Polar coordinates

Unit 4

Theory of Surfaces Introduction to space curves and surfaces, shell surfaces and characteristics, classifications of shells.

Unit 5

Introduction to Shell Theory Basic concepts of the theory, equilibrium equations in curvilinear coordinates, force displacement relations, Membrane analysis of shells of revolution and cylindrical shells under different loads.

Reference Books

1. J. N. Reddy: Theory and Analysis of Elastic Plates and Shells: CRC Press

2. H. Kraus: Thin Elastic Shells: John Wiley and Sons

3. S. Timoshenko and W. Krieger: Theory of plates and shells: Mc – Graw Hill

4. J. N. Reddy: Mechanics of Laminated Composite Plates and Shells: CRC Press

5. A. C. Ugural: Stresses in Plates and Shells: Mc Graw Hill

6. K. Chandrashekhara: Theory of Plates: Universities Press

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

First YEAR (Semester – I

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Computer Aided Analysis and Design**	**MTSE—1005(c)**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the various programming languages.

CO2: Understand the computer graphics concepts.

CO3: Analyze and design various buildings using CAD.

CO4: Analyze and design concrete and steel structures.

CO5: Analyze and design of foundations and retaining structures.

Course Contents

Unit: 1. Programming Languages :-Overview of programming languages FORTRAN-77/C++.

Unit: 2. Computer Graphics

Introduction and applications point plotting and line generation, Computer Graphics Co-ordination Systems. X-D windowing and clippling.

Unit: 3. Computer Aided Design

CAD in building design and planning,

Unit: 4 Construction Management, design and detailing of concrete and steel structures.

Unit: 5 Computer Aided Analysis and Design of foundations and retaining structures: Design of shallow foundations; Pile Foundations, Retaining Walls.

Reference Books

1. Lafore, Robert. (2006) Object oriented programming in CPP. Galgotia Publications Pvt. Ltd.

2. Balaguruswamy, E.(2011) Programming in C(5th ed.). Tata McGraw Hill Education Private Limited

3. Syal & Gupta. Computer programming and engineering analysis

4. AutoCAD, SolidEdge, Cadlab software and Manuals

5. Krishnamurthy, C.S. & Rajeev, S. & Rajaraman, A. Computer Aided Design Software and Analytical tools, (2nd ed.). Narosa Publishing House New Delhi.

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

First YEAR (Semester – I

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Rock Mechanics and Foundation Engineering**	**MTSE— 1005(D)**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the classification and failure criteria of rocks and rock masses

CO2: Estimate the strength and deformability of rock mass.

CO3: Evaluate the rock and rock masses using FEM approach

CO4: Understand the seismic consideration and stress measurement in rocks.

CO5: Estimate foundation capacity of rock mass and design a suitable foundation

Course Contents

Unit 1:

Classification of Rocks and Rock masses, Laboratory and in - situ testing of rock, insitu stresses and their measurement, Analysis and design of underground openings, Failure criteria for rock and rock masses.

Unit 2:

Strength and deformability of jointed rock mass, Stability of rock slopes, Methods to improve Rock mass properties, Rock reinforcement.

Unit 3:

FEM Approach, Numerical modeling of rocks and rock masses, Stability of rock slopes.

Unit 4:

Seismic considerations, Measurement of stress and stress in rocks, rock fracturing in compression, stress distribution in rocks

Unit 5:

Foundations on Rock: Shallow foundations, Pile and well foundations, Basement excavation, Foundation construction, Allowable bearing pressure. Tunnels: Rock stresses and deformation around tunnels, Rock support interaction, Tunnel driving methods, Design of tunnel lining

Reference Books

1. Central Board of Irrigation and Power - Manual on Rock Mechanics, 1988.

2. Billings, Structural Geology, PHI

3. E Hock, J Bray, Rock slope engineering

4. T Schebotarioti, Soil Mechanics, T MH

5. W Dunham, Foundations of structure clearance, TMH

6. Oberti and Duvalk, Rock mechanics on the design of structures in rock W. L. John Wiley.

7. Miller, Rock mechanics

8. R. E. Goodman, Introduction to Rock Mechanics, John Wiley & Sons, New York.

9. T. Ramamurthy Engineering in Rocks for Slopes, Foundation and Tunnels, Prentice Hall India Pvt. Ltd.

10. Jaeger, Cook and Zimmerman, Fundamentals of Rock Mechanics, Blackwell Publishing.

11. L. Obert and Wilbur I. Duvall, Rock mechanics and the design of structures in rock, John Wiley & Sons, Inc.

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

First YEAR (**Semester – I**

Course Content & Grade

Course Code:

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	SWAYAM/NPTEL-MOOC-1	MTSE—1005(E)/Code as per SWAYAM/NPTEL-MOOC-1	**3L-1T-0P**	**4**

Guideline

SWAYAM/NPTEL-MOOC-1 courses is a programme initiated by Government of India and designed to achieve the three cardinal principles of education via, access, equity and quality. The objective of this effort is to take the best teaching resources to all, including the most disadvantaged SWAYAM seeks to bridge divide for students who have hitherto remained untouched by the digital revolution and have not been able to join the mainstream of the knowledge economy.

Courses delivered through SWAYAM/NPTEL-MOOC courses are available free of cost to the learners, however learns wanting SWAYAM/NPTEL-MOOC courses Certificate should register for the final proctoral exams that come at a fee and attend in-person at designated center on specified dates.

In perspective of the need to create synergist between the salient features of 'anytime anywhere format of e-learning and traditional classrooms chalk and talk method to develop a unique content delivery mechanism which is responsive to learners' needs and ensuring seamless transfer of knowledge across geographical boundaries. RKDF University proposes to incorporate MOOC courses delivered by SWAYAM/NEPTEL portal. Student can opt any courses from SWAYAM/NPTEL-MOOC courses as an elective Base on following condition

1. He/She has consulted and inform to the departmental coordinator regarding courses selection and mark obtain to him. No Subjected should be repeat from courses which is offer and courses which are selected.

2. He/She can opt from only courses which are not available from University.

3. Only enrollement in the courses and learning from it is free as this is a project supported by MHRD GOI.

4. To get Certificate from SWAYAM-NPTEL. The learner has to separately register and appear for a proctored exam at a fee as decided by host institution/Mooc of Rs. 1000/- Which be bearded by Student only.

5. If student successful appear in exam and obtain certificated of courses of 12 week course to earn 3 credits(As recommended by NPTEL) to suffice for required credit for the award of his/her degree.

6. The NPTEL recommended that an additional 1 credit may be awarded given by University. This credited can be given based on attendances, assignment and tuitorial work submitted by the students.

7. All students would be motived to take courses online with SWAYAM/NPTEL every semester to better prepared for the future so that they can cultivate the habilt of self-study online. The University propose to implement MOOC courses under Department. The University also made provision if student unable to get certificate He/She still be able to appear in the department elective courses and complete there degree

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Lab – I, Advance Concrete Lab**	**MTSE—1006**	**0L-0T-6P**	**6**

Lab – I, Advance Concrete Lab

Course Outcomes

After studying this subject, the students will be able to:

CO1: Perform the various laboratory tests on cement, fine aggregates, coarse aggregates and reinforcement steel bars.

CO2: Perform the various laboratory tests on fresh and hardened concrete.

CO3: Mix Design of concrete and various special concretes.

List of Experiments:

Laboratory work shall be based on the topics of Advanced Concrete and as per the facility available in the institution.

1. To study and perform the important physical tests on various types of cement.
2. To study and perform the important physical tests on fine aggregates.
3. To study and perform the important physical tests on coarse aggregates.
4. To study and perform the physical tests on reinforcement steel bars.
5. To study and perform the various tests on fresh and hardened concrete.
6. To carry out the mix design of concrete as per IS 10262:2019 or latest version.
7. To carry out the mix design of special concretes such as:

- High strength concrete (HSC) (such as M40, M50 and so on)
- Self compacting concrete (SCC)
- Pumpable concrete
- Mass concrete
- OPC with fly ash
- OPC with GGBS
- OPC with industrial or agricultural wastes

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

First YEAR (**Semester – I**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Lab – II, CAD Lab**	**MTSE— 1007**	**0L-0T-6P**	**6**

Lab – II, **CAD Lab**

Course Outcomes

After studying this subject, the students will be able to:
CO1: Understand the various features of software dealing with analysis and design of structures
CO2: Analyze and design the various structures using structural software's.
Introduction and important features of software dealing with analysis and design of structures.
The following structures has to be performed using various analysis and design software's or any open source software's as per the facility available in the institution.
1. Analysis and design of various types of beams subjected to different loadings.
2. Analysis and design of slabs
3. Analysis and design of plane frames with different support conditions.
4. Analysis and design of multi-storey frames
3 Introduction to various softwares for structural analysis and design i.e.;

STAAD Pro, ANSYS, ETABS etc.

4 To analyse the various type of trusses using structural analysis software (Staad-Pro) and verify the results by theoretical methods

5 To analyse the different type of beams using structural analysis software (Staad-Pro) and verify the results by theoretical methods

6 To analyse the portal frame using structural analysis software (Staad-Pro) and verify the results by theoretical methods

7 Analyse of RCC building frame (regular plan configuration) for gravity load combination using structural analysis software (Staad-Pro)

8 Analyse of RCC building frame (regular plan configuration) for various load combination using structural analysis software (Staad-Pro)

9 to 12 Analysis and design of beam and other components